MANIPAL INSTITUTE OF TECHNOLOGY
MANIPAL
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III SEMESTER B.TECH MAKE UP EXAMINATION JAN- 2025

SUBJECT: ENGINEERING MATHEMATICS III [MAT 2124]

Date of Exam: **2-1-2025** Time of Exam: **9:30AM-12:30PM** Max. Marks: **50**

Q. No	Question	M
1A	Find the Fourier series expansion for the function $f(x) = \|x\|, \quad -2 < x < 2,$ having period 4. Hence deduce that $1+\frac{1}{3^2}+\frac{1}{5^2}+\frac{1}{7^2}+\cdots=\frac{\pi^2}{8}.$	**3**
1B	Show that the complex valued function $f(z) = \sin z$ is analytic everywhere on the complex plane.	**3**
1C	Find the angle between the surfaces $z = x^2 + y^2$ and $z = \left(x-\frac{\sqrt{6}}{6}\right)^2 + \left(y-\frac{\sqrt{6}}{6}\right)^2$ at the point $P = \left(\frac{\sqrt{6}}{12}, \frac{\sqrt{6}}{12}, \frac{1}{12}\right)$.	**4**
2A	Find a_0 , a_1, a_2 in the Fourier series for y, given x: 0 1 2 3 4 5 y: 4 8 15 7 6 2	**3**
2B	Evaluate the following integral $\oint_C \frac{4-3z}{z^2-z}dz$ around any simple closed (counterclockwise) curve C in each of the following cases. (i) 0 and 1 are inside C. (ii) 0 is inside and 1 is outside.	**3**
2C	Suppose $\vec{v} = \vec{w} \times \vec{r}$, where $\vec{r}$ is a position vector and $\vec{w}$ is a constant vector. Prove that $div\,\vec{v}$ =0 and $curl\ \vec{v} = 2\vec{w}$.	**4**

3A	Expand $$f(x) = \cos x, \quad 0 < x < \pi,$$ as a half range Fourier series of sine terms.	**3**
3B	If $f(z)$ is analytic, show that $$\left[\frac{\partial^2}{\partial x^2} + \frac{\partial^2}{\partial y^2}\right] |f(z)|^2 = 4\,|f'(z)|^2$$	**3**
3C	Find the Fourier sine transform of $$f(x) = \frac{1}{x}e^{-ax},$$ where a is any positive constant.	**4**
4A	Obtain the Taylor's series expansion of the function $\frac{1}{z^2-z-6}$ about $(i)\ z = -1 \quad (ii)\ z = 1,$ and hence find the region of convergence.	**3**
4B	Show that $\vec{F} = (2xy + z^3)i + x^2 j + 3xz^2 k$ is a conservative force field. (a) Find the scalar potential (b)Find the work done in moving an object in this field from (1,-2,1) to (3,1,4).	**3**
4C	Solve the partial differential equation $$\frac{\partial^2 z}{\partial x \partial y} = \sin x \sin y$$ for which $\frac{\partial z}{\partial y}(0, y) = -2 \sin y$ and $z = 0$ when y is an odd multiple of $\frac{\pi}{2}$.	**4**
5A	Verify Stoke's theorem for $\vec{F} = (2x - y)i - yz^2 j - y^2 zk$, where S is the upper half surface of the sphere $x^2 + y^2 + z^2 = 1$ and C is its boundary.	**3**
5B	Solve $$\frac{\partial u}{\partial x} = 4\frac{\partial u}{\partial y}$$ given that u(0,y) = $8e^{-3y}$ using the method of separation of variables.	**3**
5C	Evaluate $\int_c \frac{\cos z}{z^3}\, dz$, where $C: |z| = 1$.	**4**